

Does the Distribution of Trade Costs across Firms Matter?

Firm-Level Freight Costs, Selection, and Welfare

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Abstract

Quantitative trade models summarize bilateral trade costs with a single iceberg wedge common to all firms. Using Chilean customs records, I show that larger importers pay systematically lower freight rates within product, route, time, and carrier cells; controlling for shipment size attenuates this gradient by about 40 percent, and the profile rises sharply among the smallest firms. I introduce a firm level export cost schedule into the model of Arkolakis et al. (2019). Under CES, the distribution of trade costs carries no first order welfare information beyond the domestic expenditure share and the trade elasticity, and only the trade weighted mean of a firm specific shock matters. With variable markups, the distribution matters through pass-through, the choke price, and entry. Disciplined by the data and recalibrated to the same aggregate moments, freight heterogeneity visibly reshapes export selection but changes welfare modestly: in the 2021–22 freight spike, heterogeneous and comparable uniform shocks differ in welfare by less than 0.1 percentage points.

1 Introduction

Quantitative trade models usually summarize the cost of serving a foreign market with one bilateral iceberg trade cost. Every firm shipping from country i to country j faces the same proportional wedge, and firm heterogeneity enters through productivity, market access costs, or demand. This assumption is powerful for aggregation, but it rules out a simple possibility: firms using the same international route may face systematically different shipping costs. This paper asks whether that distribution of trade costs across firms matters for aggregate trade and welfare. The question became especially salient during the 2021–22 shipping disruption, when global shipping costs rose sharply and propagated into import prices and inflation (Carrière-Swallow et al., 2023). Chile provides a useful setting because its transaction level customs records report freight charges directly.

Chilean customs data show that the distribution is systematic. Larger importers pay lower freight rates. The relationship appears on the Shanghai–San Antonio route, across routes from China, and across all maritime origins. It survives product and time controls, alternative measures of firm size, predetermined size, and comparisons within carrier–route–month–year cells. In the baseline specification, a one unit increase in log firm size is associated with a 0.55 log point lower freight rate; doubling firm size therefore lowers the rate by about $0.55 \ln 2 \simeq 0.38$ log points, roughly 5 percent of the average freight rate. In the raw data, mean freight/FOB falls from 22 percent in the smallest importer size quintile to 7 percent in the largest (Appendix Table B.6). Controlling for shipment weight and value attenuates this coefficient by about 40 percent, which points to an important role for shipment characteristics. The freight bill rises less than proportionally with the size of a shipment, consistent with scale economies in shipping, although this evidence is descriptive rather than causal. A residual relationship remains, and carrier selection alone cannot explain it. The profile is also strongly nonlinear: freight rates are roughly flat over most of the size distribution and rise sharply among the smallest firms.

I introduce this empirical schedule into the framework of Arkolakis et al. (2019). Export costs take the form

$$\tau_{ij}(z) = \bar{\tau}_{ij}\phi(z), \quad j \neq i,$$

while domestic sales are frictionless. The common component $\bar{\tau}_{ij}$ captures the bilateral level of trade costs; $\phi(z)$ captures the firm level profile. I use a power law profile as an analytical benchmark and a saturating profile for the quantitative analysis. The latter is disciplined directly by the relationship between freight rates and an importer’s rank in the size distribution. The calibration is deliberately conservative: every model is recalibrated to the same import share and the same general equilibrium trade elasticity. Thus, any

remaining difference is not driven mechanically by changing the aggregate responsiveness of trade.

The first result is a sufficient statistics benchmark. Under CES demand and free entry, the profile $\phi(z)$ has no additional first order welfare effect conditional on the domestic expenditure share and the trade elasticity. More generally, for a firm specific freight shock, only its trade weighted mean matters to a first order. This is an envelope result: reallocations and selection induced by the shock are second order around the efficient CES allocation. With variable markups, the sufficient statistics result fails in a precise way. The distribution of trade costs matters through three objects: heterogeneous pass-through, the response of the choke price, and the free entry margin. The last channel is quantitatively the most important in the calibrated model because heterogeneous export costs change the profit content of foreign sales relative to domestic sales.

Quantitatively, the departure from the standard benchmark is real but modest. Holding the import share and trade elasticity fixed, the saturating freight cost profile lowers the share of active firms that export from 7.5 to 6.1 percent. The local welfare sensitivity to the domestic expenditure share rises from 0.1919 in the ACDR benchmark to 0.2067, an increase of 7.8 percent. But this local difference translates into much smaller differences for economically relevant finite changes: a 20 percent reduction in the common component of trade costs raises welfare by 2.77–2.90 percent, depending on how the freight profile is extrapolated below the observed export cutoff, rather than 2.59 percent in ACDR, while a 25 percent increase lowers welfare by 0.91 percent in both the benchmark and the saturating specification.

The 2021–22 freight spike provides an empirically relevant episode for asking a question that a homogeneous cost model cannot address: for a given aggregate increase in freight costs, does it matter which firms bear it? On the Shanghai–San Antonio route, feeding the observed rank specific increase into the model produces a welfare loss of 0.58 percent, compared with 0.60 percent under a uniform shock with the same model based trade weighted mean and 0.61 percent when the uniform shock is sized using observed FOB weights. The welfare losses are nearly identical, but exporter selection differs more visibly: the share of firms that export falls to 3.76 percent under the heterogeneous shock and to 4.02 percent under the benchmark uniform one. The result is similar across all maritime routes, and across alternative trade weighted benchmarks the welfare difference remains below 0.1 percentage points. The empirical and theoretical results therefore point to the same conclusion: the distribution of trade costs matters visibly for who trades, but surprisingly little for aggregate welfare once the aggregate size of the shock is controlled for.

The paper contributes to three literatures. First, it adds to the evidence on transportation

costs by documenting a robust within route size gradient and by separating shipment size economies from carrier selection and residual firm level advantages. Second, it extends the sufficient statistics logic of Arkolakis et al. (2012) and Arkolakis et al. (2019) to a setting in which trade costs themselves are distributed across firms, characterizing both when that distribution is irrelevant and the channels through which it matters with variable markups. Third, it disciplines that distribution directly with micro freight data rather than choosing it to improve aggregate fit. One implication is deliberately negative: empirically plausible freight heterogeneity does not explain the large dispersion of exporter sales. Under the power law Pareto benchmark, doing so would require a substantially lower effective trade elasticity.

The rest of the paper is organized as follows. Section 2 relates the paper to the literature. Section 3 describes the customs data and measurement. Section 4 documents the size gradient and examines its proximate sources. Section 5 presents the model. Section 6 disciplines the firm level export cost profile and the aggregate parameters. Section 7 reports quantitative implications for selection and markups. Section 8 derives the welfare results and studies both common and distributional freight shocks. Section 9 concludes.

2 Related Literature

This paper is closest to three strands of work. The first is the literature on heterogeneous firms, trade elasticities, and welfare sufficient statistics. Melitz (2003) provides the canonical selection framework, while Arkolakis et al. (2012) show that, across a broad class of trade models, the domestic expenditure share and the trade elasticity summarize the gains from trade. Arkolakis et al. (2019) extend this logic to variable markups and distinguish the micro elasticity faced by an individual firm from the macro elasticity governing aggregate trade. Relatedly, Atkeson and Burstein (2010) show that large firm level responses to trade cost changes need not imply large welfare effects, and Melitz and Redding (2015) emphasize that sufficient statistic formulas can fail when the trade elasticity varies with the level of trade costs. I depart from these models by allowing the iceberg cost itself to vary systematically across firms. The theoretical contribution is to ask whether the distribution of those wedges contains welfare relevant information beyond the usual sufficient statistics. Under CES it does not to a first order; with variable markups it does through pass-through, the choke price, and entry. This also connects the paper to envelope and network style sufficient statistics results such as Hulten (1978) and Baqaee and Farhi (2020), and to the efficiency result of Dhingra and Morrow (2019) under CES monopolistic competition.

The second strand studies the magnitude and structure of transportation costs. Hummels (2007) documents the importance of freight costs relative to tariffs, while Hummels and Skiba

(2004) and Hummels et al. (2009) show that shipping charges vary with product characteristics and value to weight ratios. Irarrazabal et al. (2015) quantify additive trade costs, while Kropf and Sauré (2014) and Hornok and Koren (2015) emphasize fixed costs per shipment and the endogenous choice of shipment frequency. Holmes and Singer (2018) document scale economies from container indivisibilities and consolidation. Brancaccio et al. (2020) endogenize freight rates in a spatial model of maritime shipping, and Carrière-Swallow et al. (2023) study the macroeconomic transmission of the 2021–22 shipping cost surge. Most closely, Ardelean and Lugovskyy (2023) use Chilean customs data to show that larger importers obtain lower freight rates and develop a model of carrier search and price discrimination. My focus is complementary. I do not model the industrial organization of shipping; instead I treat the observed freight schedule as a reduced form input into a quantitative heterogeneous firm model and ask whether its distribution changes selection and welfare.

A third strand allows effective trade costs to evolve or differ at the firm level. Alessandria et al. (2021), for example, study investments that gradually reduce firm level export costs. Evidence on preference utilization and rules of origin likewise shows that effective policy costs can vary across firms because access to preferential regimes is costly (Hayakawa, 2015; Cadot et al., 2014). The mechanism here is different: I discipline a cross sectional schedule of freight costs using firm size ranks and hold that schedule fixed when the common component of trade costs changes. I then separately study shocks that change the distribution of freight costs itself.

3 Data and Measurement

I use transaction level import records from the National Customs Service of Chile. The baseline sample covers maritime imports from 2007 to 2017; later years are used to measure the 2021–22 freight shock in Section 8.3. The main regressions use single product declarations so that product fixed effects have a clean interpretation. Importer identifiers change in 2018, so I exclude that year and do not link firms across the pre and post 2018 samples. The records identify the importer, origin and destination ports, country of origin, product, transaction value, freight charges, insurance, weight, mode of transport, and the carrier when available. The main cross sectional evidence uses all maritime routes, while the calibration of the firm level cost profile focuses on Shanghai–San Antonio, a large and relatively homogeneous route for which the size gradient is measured precisely.

Freight cost measure. For shipment s , I define the freight wedge as

$$\tau_s^F = 1 + \frac{\text{Freight}_s}{\text{FOB}_s}, \quad (1)$$

and use $100 \ln \tau_s^F$ as the dependent variable. This is approximately the ad valorem freight rate in percentage points and maps naturally into an iceberg cost. I exclude insurance from the baseline measure because Chilean Customs imputes insurance at 2 percent of FOB when it is not documented, and those imputed records are disproportionately concentrated among small importers. Appendix Table B.6 shows that insurance equals the imputation rate in 56.6 percent of transactions in the smallest size quintile but only 8.8 percent in the largest; the same table also documents the raw freight gradient, with mean freight/FOB of 22.0 and 7.0 percent in those two quintiles. I also drop transactions for which freight equals exactly 5 percent of FOB, the Customs imputation rate when freight is not documented. Appendix Table B.8 shows that using CIF/FOB or retaining the imputed freight observations does not change the qualitative results.

Firm size and empirical specification. The baseline size measure is the firm’s total monthly imports at FOB values, summed across all origins. Using FOB rather than CIF avoids mechanically embedding freight charges in the size measure. I estimate transaction level regressions of the form

$$100 \ln \tau_{sfprt}^F = \beta \ln \text{Size}_{ft} + X'_{sfprt} \gamma + \mu_p + \mu_r + \mu_t + \varepsilon_{sfprt}, \quad (2)$$

where s indexes shipments, f firms, p products, r routes and t month-years. Depending on the specification, X includes weight, shipment value, importing experience, and carrier relationships; fixed effects are made progressively more demanding. Standard errors are clustered by firm. The coefficient β is descriptive rather than a causal elasticity. I therefore place particular weight on specifications using size measured in the previous year or excluding the current transaction, and on within carrier–route–month-year comparisons.

4 Firm Size and Freight Costs

The central empirical fact is a negative relationship between firm size and the freight wedge. Table 1 reports the same specification for Shanghai–San Antonio, all routes from China, and all maritime origins. A one unit increase in log firm size is associated with a 0.48 to 0.55 log point lower freight rate. Since size enters in logs, doubling firm size lowers the freight rate by

$0.55 \ln 2 \simeq 0.38$ log points, about 5 percent of the average rate. This controlled relationship mirrors the raw size gradient in Appendix Table B.6.

Table 1: Firm size and freight costs across samples

	Shanghai–San Antonio	China, all routes	All origins
Firm size (log)	−0.487*** (0.075)	−0.475*** (0.032)	−0.551*** (0.028)
Experience (years importing)	−0.115* (0.066)	−0.060** (0.028)	−0.097*** (0.025)
Weight (log)	0.037 (0.071)	−0.014 (0.027)	−0.109*** (0.027)
Product FE	Yes	Yes	Yes
Month-year FE	Yes	Yes	Yes
Route FE	–	Yes	Yes
Observations	66,629	1,230,891	3,373,606
Firms	5,447	31,452	60,158
R^2	0.408	0.393	0.423

Notes: Dependent variable: $100 \ln(1 + \text{Freight}/\text{FOB})$. Maritime imports, 2007–2017, excluding transactions with imputed freight. Firm size is the firm’s monthly FOB imports from all origins. Single-product declarations. Standard errors clustered by firm. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The gradient is robust to alternative size and cost measures. Using monthly FOB imports gives -0.55 ; the coefficient is -0.49 with monthly CIF imports, -0.40 when size excludes the transaction itself, and -0.35 when size is measured with the previous year’s imports. The gradient also scales with the overall level of freight rates: the coefficient is -0.55 in 2007–17, -0.67 in 2019–24 and -0.90 during 2020–22. Relative to the average freight rate in each period, however, the gradient is remarkably stable at roughly 7–8 percent of the average rate per unit of log size. Appendix Tables B.1, B.7, B.8, and B.9 report these robustness checks.

4.1 Why do larger importers pay less?

The data point to two distinct sources of the size gradient. The first lies in the characteristics of shipments, in particular their scale. Larger importers make larger consignments, and freight grows less than proportionally with shipment size because bills of lading and port handling contain fixed components, carriers apply minimum charges, and consolidated cargo is more expensive per unit than full container loads. Table 2 examines this channel. Because the baseline already controls for weight, the change comes from adding the shipment’s FOB value, which reduces the firm size coefficient from -0.55 to -0.33 . With size measured using the previous year’s imports, the corresponding coefficient is -0.22 . Controlling for shipment

characteristics therefore attenuates the raw gradient by about 40 percent. Conditional on weight, however, shipment value also reflects value per kilogram, quality, and product composition, so this attenuation measures the role of shipment characteristics broadly rather than that of scale economies alone. The shipment coefficients are nonetheless consistent with scale economies: scaling up a shipment of the same good, so that weight and value rise together by one log unit, changes the freight rate by $3.45 - 4.58 = -1.12$ log points. The sign reversal on weight across specifications reflects the distinction between heavy, low value products and larger consignments of the same product.

Table 2: Shipment size, carrier conditions, and the size gradient

	(1)	(2)	(3)	(4)	(5)
Firm size (log)	-0.551*** (0.028)	-0.330*** (0.021)	-0.220*** (0.020)	-0.348*** (0.021)	-0.225*** (0.019)
Weight (log)	-0.109*** (0.027)	3.453*** (0.068)	3.251*** (0.070)	3.357*** (0.066)	3.169*** (0.067)
Shipment FOB value (log)		-4.575*** (0.090)	-4.354*** (0.094)	-4.449*** (0.091)	-4.241*** (0.093)
Firm size measured over	Month	Month	Prev. year	Month	Prev. year
Product FE	Yes	Yes	Yes	Yes	Yes
Route and month-year FE	Yes	Yes	Yes	—	—
Carrier $\times$ route $\times$ month-year FE	No	No	No	Yes	Yes
Observations	3,373,606	3,373,606	3,046,158	3,005,791	2,711,881
R^2	0.423	0.526	0.523	0.626	0.624

Notes: Dependent variable: $100 \ln(1 + \text{Freight}/\text{FOB})$. Single product maritime imports, 2007–2017. Column (1) is the baseline. Columns (2)–(5) control for shipment weight and FOB value. Firm size is monthly FOB imports in columns (1), (2), and (4), and previous-year FOB imports in columns (3) and (5). Columns (4)–(5) compare shipments carried by the same carrier on the same route and in the same month-year. Holding shipment composition fixed, the sum of the weight and shipment-value coefficients is the effect of scaling up a shipment proportionally. Standard errors clustered by firm.

Shipment value is also the denominator of the freight rate, so Appendix Table B.5 checks that the conditional gradient is not an artifact of this overlap or of misdeclared values. The size coefficient is the same whether the dependent variable is $\ln(\text{Freight}/\text{FOB})$ or log freight, and in the log freight regression the freight bill rises by only 0.88 percent when the weight and value of a shipment both rise by 1 percent. Freight per kilogram, a cost measure that does not use declared values, also falls with firm size. Instrumenting the shipment’s value with the firm’s other shipments, replacing the shipment’s weight and value with those of the firm’s typical shipment, or trimming extreme unit values gives size coefficients between -0.16 and -0.20 , compared with -0.22 in column (3) of Table 2.

The second source is not simple sorting into cheap carriers. Adding carrier fixed effects

leaves the size coefficient essentially unchanged (-0.551 versus -0.554), and within carrier–route–month–year cells the size coefficient is -0.77 , holding fixed the firm’s transactions with the carrier (Appendix Table B.2, column 4). Table 2 shows that even after controlling jointly for shipment weight, shipment value, and carrier–route–time conditions, the coefficient is -0.35 with current size and -0.23 with predetermined size. On Shanghai–San Antonio, carrier controls likewise leave the size gradient intact (Appendix Table B.3). Full carrier specifications show that importers using more carriers pay lower rates conditional on their number of shipments, while firms pay less to carriers with which they concentrate a larger share of shipments. These patterns are consistent with search and relationship discounts, but I do not attempt to identify a structural model of the shipping market.

An equal firm weighting exercise clarifies who obtains the residual advantage. On Shanghai–San Antonio, net of shipment weight and value, the coefficient is -0.20 when each shipment receives equal weight but only -0.06 , and statistically insignificant, when each firm–year receives equal weight (Appendix Table B.4). The remaining price advantage is therefore concentrated among frequent importers. This distinction is useful for interpretation. The quantitative model does not treat $\phi(z)$ as a primitive shipping technology. It is a reduced form schedule of effective freight costs by firm type, incorporating shipment scale and any residual relationship advantage. The object disciplined below is the observed distribution of that schedule, not a particular industrial organization mechanism.

5 Model

I extend the environment of Arkolakis et al. (2019) in one dimension: the iceberg cost of exporting varies systematically across firms. The quantitative model uses the generalized CES demand specification with a choke price; CES provides a useful benchmark because markups are constant and the allocation is efficient under free entry.

5.1 Demand, firms, and pricing

There are two symmetric countries, one factor of production, labor, and a continuum of differentiated varieties. Labor is the numeraire in the quantitative exercises. In the variable markup economy, preferences are additively separable up to a monotone transformation, with the generalized CES subutility used by ACDR. Demand for a variety with price p can be written

$$q(p; P) = \begin{cases} \alpha \left[\left(\frac{p}{P} \right)^{-\sigma} - 1 \right], & p < P, \\ 0, & p \geq P, \end{cases} \quad (3)$$

where P is the choke price. The quantitative calibration uses $\alpha = 1.5$ and $\sigma = 4$. The CES benchmark instead has constant demand elasticity and a constant markup.

Potential entrants pay f_e units of labor and draw productivity z from a Pareto distribution,

$$G(z) = 1 - \left(\frac{b}{z}\right)^\theta, \quad z \geq b. \quad (4)$$

In the generalized CES economy there are no fixed costs of serving a market; the choke price generates domestic and export cutoffs. The CES benchmark allows fixed market access costs in order to generate interior cutoffs.

A firm with marginal cost c chooses its monopoly price. Let

$$v \equiv \frac{P}{c}$$

denote relative efficiency. The optimal markup is a function $\mu(v)$, and the firm sells if and only if $v > 1$. I denote by

$$\rho(v) \equiv \frac{d \ln \mu(v)}{d \ln v} \quad (5)$$

the elasticity of the markup with respect to relative efficiency. A change in marginal cost is therefore passed through into price at rate $1 - \rho(v)$.

5.2 Firm specific export costs

The key departure from the standard model is the export cost schedule. Domestic sales carry no iceberg cost,

$$\tau_{ii}(z) = 1, \quad (6)$$

while a firm from i selling in $j \neq i$ faces

$$\tau_{ij}(z) = \bar{\tau}_{ij} \phi(z), \quad \phi'(z) \leq 0. \quad (7)$$

The common component $\bar{\tau}_{ij}$ is the bilateral level of trade costs. The profile $\phi(z)$ governs how effective export costs differ across firms.

Producing one unit requires w_i/z units of labor. Marginal costs are therefore

$$c_{ii}(z) = \frac{w_i}{z}, \quad c_{ij}(z) = \frac{w_i \bar{\tau}_{ij} \phi(z)}{z}, \quad j \neq i. \quad (8)$$

The domestic cutoff satisfies $w_i/z_d = P_i$. The export cutoff z_{ij}^* solves

$$\frac{w_i \bar{\tau}_{ij} \phi(z_{ij}^*)}{z_{ij}^*} = P_j. \quad (9)$$

Thus firm level freight heterogeneity affects selection only in foreign markets. Let X_d and X_x denote domestic and export sales per entrant and let $\Pi_m = \zeta_m X_m$ denote expected profits from market $m \in \{d, x\}$. Free entry requires $\Pi_d + \Pi_x = f_e$. Aggregate spending equals labor income, $N(X_d + X_x) = L$, so

$$N = \frac{\bar{\zeta} L}{f_e}, \quad \bar{\zeta} = \lambda_{jj} \zeta_d + (1 - \lambda_{jj}) \zeta_x, \quad \lambda_{jj} = \frac{X_d}{X_d + X_x}. \quad (10)$$

Together with optimal pricing and the cutoff conditions, these equations pin down the choke price P and the measure of entrants N for a given $\bar{\tau}$ and ϕ .

For analytical results I use the power law profile

$$\phi(z) = \left(\frac{z}{z_0} \right)^{-\delta}, \quad \delta \geq 0, \quad (11)$$

where z_0 is a normalization absorbed into $\bar{\tau}$. Under this profile,

$$c_{ij}(z) = \frac{w_i \bar{\tau}_{ij} z_0^\delta}{z^{1+\delta}}, \quad (12)$$

so productivity differences are amplified only in export costs. Relative efficiency abroad is proportional to $z^{1+\delta}$.

With Pareto productivity, aggregate exports under the power law satisfy

$$X_{ij} = K(\delta) N_i b^\theta (w_i \bar{\tau}_{ij})^{-\frac{\theta}{1+\delta}} P_j^{1+\frac{\theta}{1+\delta}}, \quad (13)$$

where $K(\delta) > 0$ collects demand terms and constants. Holding aggregate demand shifters fixed, the partial export elasticity is therefore $\theta/(1+\delta)$. Intuitively, the profile does not create a new response of a given firm's demand to a proportional change in the common cost component; instead it changes the cross sectional distribution of relative efficiency and compresses the export extensive margin. Under Pareto productivity, the effective export efficiency $z^{1+\delta}$ is itself Pareto with shape $\theta/(1+\delta)$. The partial elasticity $\theta/(1+\delta)$ is not, in general, the equilibrium trade elasticity because P , entry, and domestic sales also respond.

The object used throughout the quantitative analysis is

$$\varepsilon_T \equiv -\frac{\partial \ln[(1 - \lambda_{jj})/\lambda_{jj}]}{\partial \ln \bar{\tau}_{ij}}, \quad (14)$$

where λ_{jj} is the domestic expenditure share.

The power law profile maps directly into firm ranks. Conditional on exporting, the share of firms at least as productive as z is

$$R = \left(\frac{z}{z_{ij}^*} \right)^{-\theta}. \quad (15)$$

Since sales increase with productivity, R is also the firm's rank in the size distribution of exporters, with R close to zero for the largest firms and close to one for the smallest. Conditional on exporting, R is uniform on $(0, 1)$, so the rank mapping does not require the model to match the dispersion of firm size. Substituting $z = z_{ij}^* R^{-1/\theta}$ into (11) gives

$$\ln \tau_{ij}(z) = \ln \tau_{ij}(z_{ij}^*) + \frac{\delta}{\theta} \ln R. \quad (16)$$

The slope of the log iceberg cost on the log size rank therefore identifies δ/θ .

6 Calibration

The calibration separates the shape of firm level freight costs from the aggregate level of trade costs. The customs data discipline the profile $\phi(z)$ through size ranks; aggregate moments discipline θ and $\bar{\tau}$.

6.1 The profile of export costs

From importers to exporters. The customs data identify the size of the Chilean importer, not the productivity of the foreign producer. I interpret the observed gradient as a reduced form relationship level trade cost: larger trading relationships obtain lower freight rates, in part because they ship larger consignments. The model assigns this schedule to exporter productivity because productivity is the source of firm size in the model. The schedule is therefore treated as a characteristic of firm type, not as a structural shipping technology.

Power law benchmark. On Shanghai–San Antonio, I rank importers each year by total FOB imports from all origins. The rank R is the share of firms on the route that year whose imports are at least as large. Regressing $\ln(1 + \text{Freight}/\text{FOB})$ on $\ln R$ with product by year fixed effects and equal firm-year weights gives

$$r = 0.0065 \quad (\text{s.e. } 0.0009),$$

using 22,999 observations and 5,470 firms. Equation (16) implies $\delta = r\theta$. Once θ is recalibrated, $\delta = 0.034$, with a 95 percent confidence interval $[0.024, 0.043]$ obtained by recalibrating at the endpoints of the confidence interval for r . A firm twice as productive as another therefore faces an iceberg cost about 2.3 percent lower under the power law benchmark.

Saturating quantitative profile. The power law is useful analytically but does not fit the shape of the freight cost profile well. Figure 1 plots the residual of $\ln(1 + \text{Freight}/\text{FOB})$, net of product by year fixed effects, in 20 bins of R . Freight rates are approximately flat for most firms and rise sharply near the bottom of the size distribution. The curve

$$\ln \tau = c + \ln(1 + \kappa R^\psi) \tag{17}$$

fits the bins with $\kappa = 0.044$ and $\psi = 8.08$ ($R^2 = 0.92$, compared with 0.28 for the log-linear profile). I therefore use

$$\phi(z) = 1 + \kappa \left(\frac{z}{z_{\text{ref}}} \right)^{-\eta} \tag{18}$$

as the quantitative baseline. In the baseline equilibrium z_{ref} is the export cutoff, so ranks identify κ and $\psi = \eta/\theta$. I keep z_{ref} fixed in counterfactuals. As z grows, $\phi(z) \rightarrow 1$, so $\bar{\tau}$ is the trade cost faced by the largest exporters and the smallest baseline exporters pay a surcharge of about 4.4 percent.

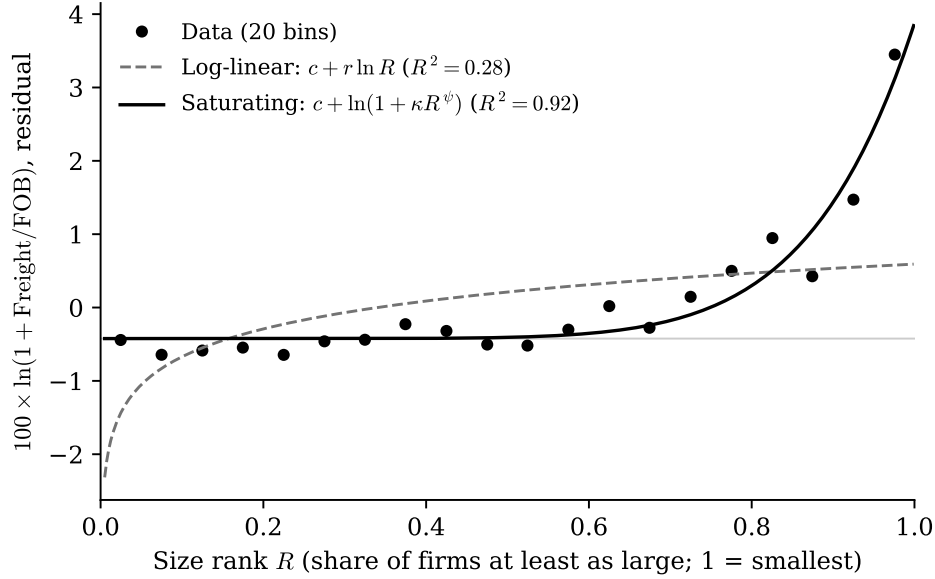


Figure 1: Freight rates by size rank, Shanghai–San Antonio, 2007–2017

Notes: Average residual of $100 \ln(1 + \text{Freight}/\text{FOB})$, net of product by year fixed effects, in 20 bins of the size rank R . Each firm-year receives equal weight. The dashed line is the least squares log-linear fit to the 20 bin means, with slope 0.0055; the regression on the underlying observations gives $r = 0.0065$. The solid line is $c + \ln(1 + \kappa R^\psi)$ with $\kappa = 0.044$ and $\psi = 8.08$. Slopes and fitted curves are in units of $\ln(1 + \text{Freight}/\text{FOB})$; the figure plots 100 times these values.

6.2 Aggregate targets

I use the generalized CES demand parameters from the ACDR working paper calibration, $\alpha = 1.5$ and $\sigma = 4$. This choice keeps the quantitative exercise internally consistent with the French exporter sales moments reported for that calibration, which come from the firm level data of Eaton et al. (2011); the published version uses a later demand calibration but does not report the corresponding firm sales distribution used below as an auxiliary validation exercise.

For each export cost profile, I jointly choose the Pareto shape θ and the common trade cost component $\bar{\tau}$ to match the same two macro moments: a general equilibrium trade elasticity $\varepsilon_T = 5$ and an imports to expenditure ratio of 6.95 percent, so $\lambda_{jj} = 0.9305$. The lower bound b is chosen to preserve ACDR's baseline ratio $z_d/b = 1.02$; this condition fixes the fraction of entrants that produce and, conditional on the other targets, does not affect the outcomes reported below. Given θ , the micro profile then pins down $\delta = r\theta$ in the power law economy and $\eta = \psi\theta$ in the saturating economy. Table 3 reports the resulting parameters.

Table 3: Calibration (generalized CES)

	Constant costs (ACDR)	Power law profile $\phi = (z/z_0)^{-\delta}$	Saturating profile $\phi = 1 + \kappa(z/z_{\text{ref}})^{-\eta}$
<i>Common targets and demand parameters</i>			
Trade elasticity, ε_T	5	5	5
Imports/expenditure (%)	6.95	6.95	6.95
z_d/b	1.02	1.02	1.02
α, σ	1.5, 4	1.5, 4	1.5, 4
<i>Firm-level profile</i>			
Micro moment	—	$r = 0.0065$	$\kappa = 0.044, \psi = 8.08$
<i>Calibrated parameters</i>			
θ	5.000	5.159	5.579
$\bar{\tau}$	1.680	1.606	1.584
δ	0	0.034	—
κ, η	—	—	0.044, 45.1
b	0.180	0.164	0.130
z_{ref}	—	—	0.220

Notes: All specifications match the same aggregate openness and general equilibrium trade elasticity. Changes in the common level of trade costs scale $\bar{\tau}$ while the firm level profile parameters remain fixed. The power law profile uses $z_0 = 1$.

7 Quantitative Implications

Because all three economies match the same import share and trade elasticity, differences across them isolate the effect of how export costs are distributed across firms. Table 4 summarizes the main equilibrium implications.

Selection into exporting becomes stronger. The fraction of active firms that export falls from 7.5 percent under constant costs to 7.0 percent under the power law and 6.1 percent under the saturating profile. The levels of z_d and z_x are not comparable across calibrations because the productivity normalization and Pareto lower bound change. The relative cutoff z_x/z_d , by contrast, is comparable and remains close to the iceberg cost of ACDR’s marginal exporter. The chain is important. At a given θ , the firm specific profile dampens the partial export response relative to constant costs (under the power law, from θ to $\theta/(1 + \delta)$). To keep the general equilibrium trade elasticity at 5, the calibration therefore raises θ . Since z_x/z_d changes little, the higher θ thins the upper tail and lowers the fraction of active firms that export.

Markups move little. The function $\mu(v)$ is identical across the three economies, so differences in mean markups and in the sales weighted markup elasticities ρ_d and ρ_x are entirely compositional. What matters more for welfare is the profit content of sales. With heterogeneous export costs, profits are a slightly larger share of export sales than of domestic sales,

$\zeta_x > \zeta_d$. That small wedge is what activates the entry channel in Section 8.

Table 4: Equilibrium outcomes (generalized CES)

	Constant costs (ACDR)	Power-law profile	Saturating profile
Fraction of active firms that export	0.075	0.070	0.061
Relative export cutoff, z_x/z_d	1.680	1.677	1.653
Measure of entrants, N	0.167	0.163	0.152
Mean log markup, export sales	0.082	0.083	0.091
Markup elasticity, ρ_d	0.244	0.255	0.280
Markup elasticity, ρ_x	0.244	0.243	0.274
Profit share, ζ_d	0.167	0.162	0.152
Profit share, ζ_x	0.167	0.167	0.156
Consumer surplus/spending, S_d	0.151	0.145	0.129
Consumer surplus/spending, S_x	0.151	0.152	0.133
$d \ln N / d \ln \lambda_{jj}$	0.00	-0.03	-0.11

Notes: Parameters in Table 3. ρ_d and ρ_x are sales weighted markup elasticities for domestic and export sales. The profit shares ζ_d and ζ_x , consumer surplus ratios S_d and S_x , and the entry elasticity are the objects entering Proposition 2.

As an auxiliary validation, I also compare the distribution of exporter sales with the French moments reported by ACDR. The result is informative precisely because it is not a success of the mechanism: empirically plausible freight heterogeneity does not explain the observed dispersion of exporter sales. Under the power law profile, relative efficiency among exporters is Pareto with shape $\theta/(1+\delta) = 4.99$, essentially ACDR's value of 5, so the normalized sales distribution is almost unchanged. The same ratio $\theta/(1+\delta)$ is the partial trade elasticity in (13), so the dispersion of sales and the response of trade are tied together. The saturating profile compresses the bottom because its surcharge disappears quickly just above the export cutoff. Appendix C reports the corresponding percentiles. Thus the freight cost profile is disciplined by freight data, not chosen to improve the firm sales distribution. For illustration, a calibration that ignores the freight evidence and sets $\theta = 5$ and $\delta = 0.59$ brings the lower half of the French sales distribution much closer to the data, but only by lowering the effective Pareto shape to $\theta/(1+\delta) = 3.1$. Its general equilibrium trade elasticity is about 3.3 rather than 5, and the implied productivity gradient of export costs is roughly 17 times the estimate in Section 6.1. Within the power law Pareto benchmark, substantially increasing sales dispersion therefore requires giving up the aggregate trade elasticity target or imposing far more micro heterogeneity than the freight data support.

8 Welfare and Distributional Shocks

This section asks when the distribution of trade costs across firms contains welfare relevant information beyond the standard sufficient statistics. I first characterize proportional changes in the common component $\bar{\tau}$ and then turn to shocks that change the distribution itself. Proofs are in Appendix A.

8.1 When does the profile matter?

Throughout, W denotes welfare per capita measured as an equivalent variation and ε_T is the general equilibrium trade elasticity in (14).

Proposition 1 (CES demand). *Consider CES demand, free entry, fixed costs of serving each market (possibly zero), and a proportional change in $\bar{\tau}_{ij}$ with $\phi(z)$ held fixed. For any profile ϕ and any productivity distribution, to a first order,*

$$d \ln W = -(1 - \lambda_{jj}) d \ln \bar{\tau}_{ij} = -\frac{1}{\varepsilon_T} d \ln \lambda_{jj}. \quad (19)$$

If, in addition, productivity is Pareto with shape θ , $\phi(z) = (z/z_0)^{-\delta}$, the domestic and export cutoffs are interior, and the relevant sales moments are finite, then

$$\varepsilon_T = \frac{\theta [1 + \delta(1 - \lambda_{jj})]}{1 + \delta}. \quad (20)$$

The result is an envelope theorem. Conditional on the domestic expenditure share and the equilibrium trade elasticity, the firm level shape of export costs contains no additional first order welfare information under CES. The profile does change the trade elasticity itself: by (20), under the power law ε_T lies between the partial elasticity $\theta/(1 + \delta)$ and θ . This is why every specification is recalibrated to the same ε_T . The result is also local: for a finite change, the path of ε_T can depend on the profile.

With variable markups, the distribution matters beyond λ_{jj} and ε_T .

Proposition 2 (Variable markups). *Consider additively separable preferences with a choke price, as in the generalized CES specification, free entry, no fixed costs of serving a market, and a proportional change in $\bar{\tau}_{ij}$ with $\phi(z)$ fixed. For any profile ϕ and any productivity distribution, to a first order $d \ln W = -\Omega d \ln \lambda_{jj}$, where*

$$\Omega = \frac{1 - \rho_x}{\varepsilon_T} + \frac{\bar{\rho}}{(1 - \lambda_{jj})\varepsilon_T} \frac{d \ln P}{d \ln \bar{\tau}_{ij}} - \bar{S} \frac{d \ln N}{d \ln \lambda_{jj}}. \quad (21)$$

Here ρ_d and ρ_x are the sales weighted averages of the firm level markup elasticity $\rho(v)$ for domestic and export sales, $\bar{\rho} = \lambda_{jj}\rho_d + (1 - \lambda_{jj})\rho_x$, and $\bar{S}$ is consumer surplus relative to spending, aggregated across domestic and imported varieties. Moreover,

$$N = \frac{\bar{\zeta}L}{f_e}, \quad \frac{d \ln P}{d \ln \bar{\tau}_{ij}} = (1 - \lambda_{jj})(1 - \zeta_x), \quad (22)$$

where ζ_d and ζ_x are profit shares in domestic and export sales and $\bar{\zeta} = \lambda_{jj}\zeta_d + (1 - \lambda_{jj})\zeta_x$.

Equation (21) separates three channels. The first is the direct effect of trade costs on import prices, net of markup adjustment. The second captures the response of the choke price and hence markups. The third is entry: changes in the profit content of domestic and foreign sales alter free entry and the number of available varieties. In the ACDR Pareto benchmark, $\zeta_d = \zeta_x$, so this last margin is absent. Productivity dependent export costs create a wedge between the two profit shares and activate it.

For the power law profile the objects in Proposition 2 have closed forms.

Corollary 1 (Power law profile). *Let $\phi(z) = (z/z_0)^{-\delta}$ and productivity be Pareto with shape θ and lower bound b below both cutoffs, with $\theta/(1 + \delta) > \sigma - 1$ so that aggregate sales are finite. Then*

$$\zeta_d = \frac{1}{1 + \theta}, \quad \zeta_x = \frac{1 + \delta}{1 + \theta + \delta},$$

$$\frac{d \ln N}{d \ln \lambda_{jj}} = -\frac{\lambda_{jj}\theta\delta}{1 + \theta + \delta + (1 - \lambda_{jj})\theta\delta}, \quad (23)$$

$$\frac{d \ln P}{d \ln \bar{\tau}_{ij}} = \frac{(1 - \lambda_{jj})\theta}{1 + \theta + \delta}, \quad (24)$$

$$\varepsilon_T = \frac{\theta}{1 + \delta} + \frac{\theta\delta}{1 + \delta} \frac{d \ln P}{d \ln \bar{\tau}_{ij}}. \quad (25)$$

Consumer surplus relative to spending satisfies

$$S_d = \frac{1 - \rho_d}{\theta}, \quad S_x = \frac{(1 + \delta)(1 - \rho_x)}{\theta}.$$

When $\delta = 0$, $\rho_d = \rho_x \equiv \rho$, $d \ln N / d \ln \lambda_{jj} = 0$, and the expressions collapse to the ACDR benchmark,

$$\Omega_{ACDR} = \frac{1 - \rho/(1 + \theta)}{\theta}. \quad (26)$$

8.2 Quantitative welfare

Table 5 shows the main quantitative result. Because every specification matches the same import share and $\varepsilon_T = 5$, ACR would assign the same local welfare sensitivity, $1/\varepsilon_T = 0.2$, to all three. Generalized CES changes that number slightly even with constant trade costs, as in ACDR: with $\rho = 0.244$ and $\theta = 5$, equation (26) gives $\Omega = (1 - 0.244/6)/5 = 0.1919$, the value computed from exact welfare changes. The power law profile raises the local sensitivity by 2.9 percent relative to ACDR; the saturating profile raises it by 7.8 percent. The decomposition shows that entry accounts for most of the additional effect. The direct and markup channels move in opposite directions and largely offset one another.

The entry channel is larger under the saturating profile because the profit share of export sales changes endogenously as the export cutoff moves through the steep part of the freight cost schedule. As trade costs fall, more exporters operate where the premium changes sharply, raising ζ_x ; this endogenous movement accounts for roughly four fifths of the increase in the aggregate profit share $\bar{\zeta}$ and helps explain why $d \ln N / d \ln \lambda_{jj}$ is -0.11 , compared with about -0.03 under the power law.

The finite counterfactuals put the local differences in perspective. A 20 percent reduction in $\bar{\tau}$ raises welfare by 2.59 percent under constant costs and 2.66 percent under the power law, and a 25 percent increase lowers it by about 0.9 percent in all specifications. Under the saturating profile the gain is 2.90 percent, but this number depends on how the profile is extrapolated. A reduction in $\bar{\tau}$ brings in exporters below z_{ref} , a range of productivity that the baseline data do not cover. With the full extrapolation, the saturating profile makes their export costs rise steeply. This dampens the extensive margin; imports rise to 16.7 percent of expenditure, against 18.6 percent in ACDR, and the arc trade elasticity is 4.4. The larger welfare gain therefore cannot be attributed to a stronger trade response. Instead, the profit content of export sales rises and the measure of entrants increases by 2.9 percent. Capping the surcharge of these firms at its largest observed value, $(1 + \kappa)\bar{\tau}$, raises imports to 19.1 percent of expenditure, brings the arc trade elasticity to 5.2, and reduces the entry response to 0.9 percent; the welfare gain falls to 2.77 percent. Integrating the three terms of Proposition 2 along the path of $\bar{\tau}$, entry accounts for more than the entire additional gain with the full extrapolation and for about two thirds of it with the bounded extrapolation; in the latter case, the rest comes from a larger procompetitive fall in the choke price, to which the stronger import response contributes.¹ An increase in $\bar{\tau}$ instead moves the export cutoff

¹Integrating $d \ln W = -(1 - \lambda_{jj})(1 - \rho_x) d \ln \bar{\tau} - \bar{\rho} d \ln P + \bar{S} d \ln N$ from $\bar{\tau}$ to $0.8\bar{\tau}$, the direct, choke price, and entry contributions (log points) are (2.02, 0.54, 0) under constant costs, (1.90, 0.58, 0.38) with full extrapolation, and (1.99, 0.63, 0.12) with bounded extrapolation. The sums match exact welfare changes within 0.01 log points. The direct term is smaller than in ACDR because of greater markup absorption (higher ρ_x) and, under full extrapolation, a smaller import response.

above z_{ref} , where the premium is negligible: the saturating economy then behaves close to one with constant costs (the arc trade elasticity is 5.4 and entry falls by 0.2 percent), and the loss is essentially the same as in ACDR. The bounded extrapolation is therefore the conservative benchmark for liberalization.

Table 5: Welfare with alternative export cost profiles

	Ω	Direct	Markup	Entry	$\bar{\tau} \times 0.8$	$\bar{\tau} \times 1.25$
		decomposition of Ω			welfare change (%)	
Constant costs (ACDR)	0.1919	0.151	0.041	0.000	2.59	-0.91
Power-law profile	0.1974	0.151	0.042	0.004	2.66	-0.94
Saturating profile	0.2067	0.145	0.047	0.014	2.90	-0.91
bounded extrapolation ^a	0.2066	0.145	0.047	0.014	2.77	-0.91

Notes: $\Omega = -d \ln W / d \ln \lambda_{jj}$ is computed from exact equivalent variation changes of ± 0.1 percent in $\bar{\tau}$. The three specifications have the same baseline import share and trade elasticity. The last two columns report exact equivalent variation for finite changes in $\bar{\tau}$, holding the firm level profile fixed. ^a For firms below the baseline z_{ref} , the surcharge is capped at its largest baseline value, $1 + \kappa$; the local statistics are essentially unchanged because the cap lies outside the baseline support of exporters.

Appendix C shows that the additional effect stays modest under alternative rank measures, bin definitions, freight measures and weights, between 1.9 and 8.3 percent of the ACDR value of Ω , and that it largely disappears once shipment size is netted out.

8.3 Shocks to the distribution of freight costs

The counterfactuals so far scale the common component $\bar{\tau}_{ij}$ and hold the profile fixed. Freight shocks need not be uniform across firms, and the model can answer a question that a model with homogeneous costs cannot: for a given increase in freight costs, does it matter who bears it?

Proposition 3 (Heterogeneous shocks). *Consider a change in export iceberg costs, $d \ln \tau_{ij}(z)$, that may differ across firms. Let E_s and E_c denote averages across exporters weighted by export sales and by variable export costs. Under CES, to a first order,*

$$d \ln W = -(1 - \lambda_{jj}) E_s[d \ln \tau_{ij}(z)]. \quad (27)$$

Under the assumptions of Proposition 2, to a first order,

$$d \ln W = -(1 - \lambda_{jj}) E_s[(1 - \rho(v)) d \ln \tau_{ij}(z)] - \bar{\rho} d \ln P + \bar{S} d \ln N, \quad (28)$$

$$d \ln P = (1 - \lambda_{jj})(1 - \zeta_x) E_c[d \ln \tau_{ij}(z)], \quad (29)$$

and $N = \bar{\zeta} L / f_e$.

Under CES, the first order welfare effect of a freight shock depends only on its trade weighted mean: two shocks with the same trade weighted mean have the same first order effect, however differently they treat small and large firms. This is a version of Hulten’s (1978) theorem, which applies because the CES allocation is efficient. With variable markups the allocation is not efficient and, as in Baqaee and Farhi (2020), the distribution of the shock matters through three channels. A firm close to the choke price absorbs a larger share of a cost increase in its markup, so a shock concentrated among small exporters has lower pass-through. The choke price responds to a variable cost weighted rather than a sales weighted mean of the shock. Finally, the shock changes the profit content of export sales and, through free entry, the measure of entrants.

The 2021–22 freight spike provides an empirically relevant episode with which to discipline such a shock. On Shanghai–San Antonio, the average of $100 \ln(1 + \text{Freight}/\text{FOB})$ rose from 6.5 in 2019 to 18.8 in 2021–22. Figure 2 shows the change by size rank: about 10.5 log points for the largest 15 percent of importers and 12.6 for the rest. I feed a log-linear fit of that rank specific change into the calibrated saturating economy, holding the change at the top bin value for larger firms, and compare it with uniform increases in $\bar{\tau}$ of the same aggregate size. The benchmark uniform shock (U) uses the model’s baseline export sales weights, as in Proposition 3, and has a mean increase of 11.4 log points.² Because the model is symmetric, the shock raises export costs in both countries, and it holds fixed the demand and supply disruptions that accompanied the freight spike. The exercise therefore asks whether the distribution of a given freight shock across firms matters, not what the 2021–22 episode cost Chile.

Table 6 reports the results. The heterogeneous and benchmark uniform shocks generate very similar welfare losses, 0.575 and 0.602 percent, even though their effects on exporter selection differ more visibly. The share of active firms that export falls from 6.05 to 3.76 percent under the heterogeneous shock and to 4.02 percent under the uniform one, while imports fall to 3.87 and 3.90 percent of expenditure. The welfare difference is only 0.027 percentage points. A first order decomposition of that difference attributes about two thirds to entry and about one third to incomplete pass-through among small exporters. Table A.1 in the appendix verifies Proposition 3 numerically: as the shock is scaled down, the generalized CES welfare gap converges to its first order value, whereas under CES it vanishes.

The conclusion is not driven by the model’s trade weights. Weighting the rank bin changes by their observed FOB shares in 2019 gives an average increase of 11.7 log points on Shanghai–San Antonio; a uniform shock of that size lowers welfare by 0.611 percent,

²Fitting the saturating curve year by year is unstable in 2021–22, when the profile is closer to log-linear, so I use the change by rank bin directly.

again slightly more than the heterogeneous shock. The route’s aggregate freight rate, $\ln(1 + \sum \text{Freight} / \sum \text{FOB})$, rises by only 9.7 log points, but that statistic is not another weighting of the same rank specific shock: it also incorporates within bin heterogeneity and the post-shock reallocation of imports toward larger importers.³ Against a uniform shock sized by the aggregate freight rate, the heterogeneous shock is slightly more costly. Thus the sign of a difference of only a few hundredths of a percentage point can depend on how the aggregate shock is defined, but it is robust to alternative baseline trade weights.

The same pattern appears across all maritime routes. Freight rates rose less and the rank gradient was flatter: about 7.6 log points for the largest 15 percent of importers and 8.4 for the rest. The heterogeneous shock lowers welfare by 0.443 percent, compared with 0.455 percent under the uniform shock using model trade weights and 0.464 percent using observed FOB weights. The exporter share falls to 4.63 percent, compared with 4.76 and 4.72 percent under those uniform benchmarks. The qualitative result for selection is therefore robust: the heterogeneous shock leaves fewer exporters than every uniform shock reported in Panels A and B.

Panel C pushes the comparison further. Extending the fitted gradient to the very largest firms or measuring changes within bins using FOB weighted freight rates increases the welfare gap, but it remains only 0.04–0.06 percentage points on Shanghai–San Antonio. Measuring the shock from 2007–17 makes it nearly uniform and eliminates the gap. Scaling freight rates proportionally produces a welfare difference of only 0.003 percentage points. Doubling the small firm premium while holding the trade weighted mean almost unchanged moves the exporter share from 6.05 to 5.71 percent but changes welfare by less than 0.01 percent. Across all specifications, the welfare costs of heterogeneous and comparable uniform shocks differ by less than 0.1 percentage points, against total losses of roughly 0.3–0.6 percent. The distribution of a freight shock across firms therefore matters mainly for who exports: conditional on its aggregate size, its welfare cost is largely summarized by trade weighted exposure even with variable markups.

³Aggregating freight and FOB within each bin, using 2019 FOB shares across bins, gives 10.8 log points; reallocation accounts for the rest. Because the aggregate rate uses post-shock trade, it mixes the response of trade into the measured size of the shock. When the heterogeneous shock itself is measured with the FOB weighted change within each bin, its welfare cost is below that of a uniform shock sized by the aggregate rate.

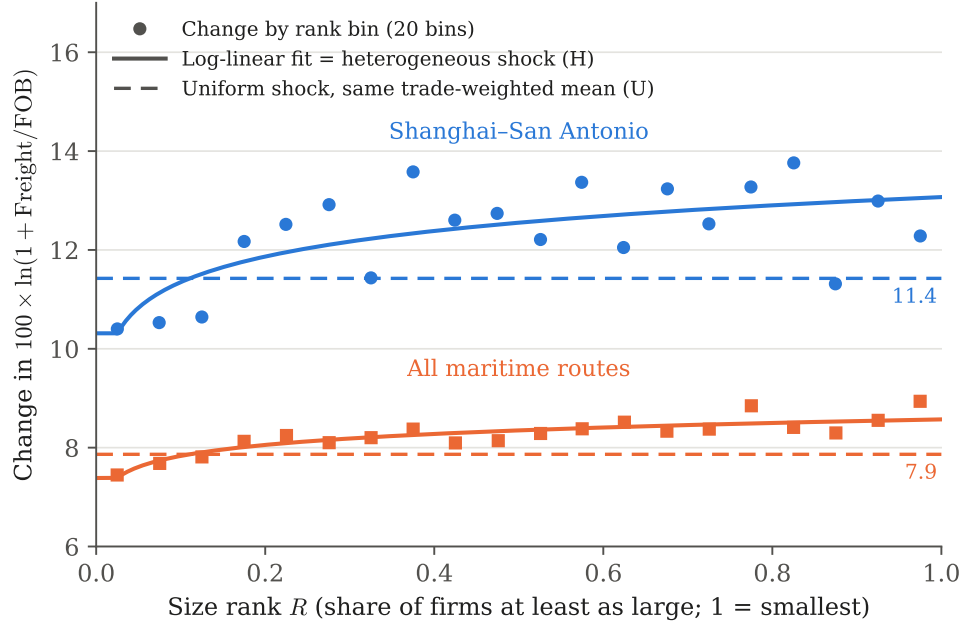


Figure 2: Change in freight rates by size rank, 2019 to 2021–22

Notes: Change in the average of $100 \ln(1 + \text{Freight}/\text{FOB})$ in 20 bins of the size rank R between 2019 and the average of 2021 and 2022, on Shanghai–San Antonio and on all maritime routes (ranks within route and year). Solid lines are least squares fits of $a + b \ln R$, held at their top bin value for larger firms; they are the heterogeneous shocks in Table 6. Dashed lines are uniform shocks with the same trade-weighted mean across the model’s baseline exporters.

Table 6: Heterogeneous and uniform freight shocks (generalized CES)

	Shock (log pts.)	Welfare (%)	Imports (% of exp.)	Exporters (% of firms)	Entrants (% change)
Baseline	–	–	6.95	6.05	–
<i>A. Shanghai–San Antonio, 2019 to 2021–22</i>					
Heterogeneous (H)	10.3–13.1	–0.575	3.87	3.76	–0.07
Uniform, model trade weights (U)	11.4	–0.602	3.90	4.02	–0.16
Uniform, data FOB weights	11.7	–0.611	3.85	3.97	–0.16
Uniform, aggregate freight rate ^a	9.7	–0.532	4.29	4.39	–0.16
Uniform, firm average	12.3	–0.635	3.72	3.83	–0.16
<i>B. All maritime routes, 2019 to 2021–22</i>					
Heterogeneous (H)	7.4–8.6	–0.443	4.68	4.63	–0.10
Uniform, model trade weights (U)	7.9	–0.455	4.71	4.76	–0.15
Uniform, data FOB weights	8.1	–0.464	4.66	4.72	–0.15
Uniform, aggregate freight rate ^a	5.6	–0.346	5.29	5.19	–0.13
Uniform, firm average	8.3	–0.473	4.62	4.68	–0.15
	Mean shock (log pts.)	H minus U (percentage points)			
		Welfare	Imports	Exporters	Entrants
<i>C. Heterogeneous minus uniform shock with the same trade weighted mean (U)</i>					
Shanghai–San Antonio, as in A	11.4	0.027	–0.04	–0.27	0.09
extrapolated to the largest firms	11.0	0.043	–0.06	–0.35	0.14
FOB weighted change within bins	10.1	0.061	–0.09	–0.58	0.21
from 2007–17 to 2021–22	10.6	–0.006	0.01	0.06	–0.02
proportional freight rates	9.9	0.003	0.01	–0.06	0.01
small firm premium doubles ^b	0.05	0.008	–0.05	–0.34	0.05
All maritime routes, as in B	7.9	0.012	–0.03	–0.13	0.05
extrapolated to the largest firms	7.7	0.018	–0.05	–0.16	0.07
FOB weighted change within bins	5.4	0.017	–0.06	–0.16	0.07

Notes: Calibrated economy with the saturating profile (Table 3). The heterogeneous shock changes the iceberg cost of each baseline exporter by the change in $\ln(1 + \text{Freight}/\text{FOB})$ observed at its size rank between 2019 and 2021–22 (Figure 2; log-linear fit, held at its top bin value for larger firms); the shock column gives its range. Panel B pools all maritime routes, with size ranks computed within route and year. Uniform shocks change $\bar{\tau}$ by a mean of the same rank specific change, weighted by the export sales of the model’s baseline exporters (U), by the FOB shares of the rank bins in 2019, or equally across firms. Welfare is exact equivalent variation. Panel C reports differences between H and U in percentage points (entrants: difference in the percentage change in N). “Extrapolated” extends the fit to firms larger than those in the top bin; “FOB weighted change within bins” measures each bin as the change in $\ln(1 + \text{bin freight}/\text{bin FOB})$; “from 2007–17” measures the change from the calibration period; “proportional freight rates” multiplies freight/FOB of every firm by the factor that matches the mean increase. ^a Change in $\ln(1 + \text{total freight}/\text{total FOB})$; besides weights, it incorporates within bin heterogeneity and reallocation across importers between 2019 and 2021–22. ^b κ doubles for every exporter.

9 Conclusion

This paper asks whether the distribution of trade costs across firms matters once standard aggregate trade moments are known. The evidence shows that the distribution is systematic. In Chilean customs data, larger importers pay lower freight rates within narrowly defined trade environments. Shipment characteristics account for an important part of the observed gradient, simple carrier sorting does not, and the profile is highly nonlinear: most of the additional freight burden is concentrated among the smallest firms.

The theory separates when that distribution matters from when it does not. Under CES, the domestic expenditure share and the trade elasticity remain sufficient statistics for proportional changes in the common level of trade costs, and the trade weighted mean is sufficient to a first order for shocks that differ across firms. With variable markups, the distribution carries additional information through pass-through, the choke price, and endogenous entry. Quantitatively, entry is the main departure from the ACDR benchmark because firm specific export costs change the profit content of foreign sales relative to domestic sales.

These departures are modest once the profile is disciplined by micro freight data and every model is calibrated to the same openness and trade elasticity. The saturating profile raises the local welfare sensitivity to the domestic expenditure share by 7.8 percent relative to ACDR, but a 20 percent reduction in the common trade cost component raises welfare by 2.77–2.90 rather than 2.59 percent, depending on the extrapolation of the freight profile below the observed cutoff, and a 25 percent increase produces essentially the same welfare loss. In the 2021–22 freight spike, heterogeneous and comparable uniform shocks differ in welfare by less than 0.1 percentage points across alternative trade weights and across all maritime routes, while the effect on exporter selection is more visible. Empirically plausible freight heterogeneity also does not explain the large dispersion of exporter sales: within the power law Pareto benchmark, generating substantially more dispersion requires a lower effective trade elasticity.

The broader implication is that firm level heterogeneity in trade costs is economically visible without overturning aggregate sufficient statistics results. It is large enough to reshape selection and the composition of trading firms, yet aggregate welfare is largely summarized by trade weighted exposure. At the same time, the theory characterizes the precise exception: with variable markups and endogenous entry, who bears a trade cost shock contains additional first order welfare information. This distinction helps clarify when micro data on the distribution of trade costs are essential for quantitative trade policy and when aggregate moments remain sufficient.

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A Proofs of the welfare results

Common setup. Countries are symmetric, and a change in trade costs is a proportional change in $\bar{\tau} \equiv \bar{\tau}_{ij} = \bar{\tau}_{ji}$ with ϕ fixed, so the equilibrium stays symmetric. The wage is the numeraire, $w_i = w_j = 1$; each country has L consumers, each supplying one unit of labor; entry requires f_e units of labor; and entrants draw productivity from a distribution G with density g . I write $\lambda \equiv \lambda_{jj}$, N for the measure of entrants in each country, $z_x \equiv z_{ij}^*$ for the export cutoff, $\tau_x(z) \equiv \bar{\tau}\phi(z)$ for the export iceberg cost, and X_d and X_x for the domestic and export sales of the average entrant. Free entry drives aggregate profits net of entry costs to zero, so aggregate income equals L and

$$N(X_d + X_x) = L. \quad (\text{A.1})$$

By symmetry, imports equal the exports of the foreign country, so $\lambda = X_d/(X_d + X_x)$. Since $d \ln[(1 - \lambda)/\lambda] = -d \ln \lambda / (1 - \lambda)$, the definition of ε_T gives

$$d \ln \lambda = (1 - \lambda) \varepsilon_T d \ln \bar{\tau}. \quad (\text{A.2})$$

A.1 Proof of Proposition 1

Step 1: sales. With CES demand with elasticity $\sigma > 1$, firms charge $p = \sigma c / (\sigma - 1)$, and a variety with price p sells $(p/\mathcal{P})^{1-\sigma} L$, where $\mathcal{P}$ is the CES price index, common to both countries. Let $A \equiv L[(\sigma - 1)\mathcal{P}/\sigma]^{\sigma-1}$. A firm with productivity z sells

$$r_d(z) = A z^{\sigma-1} \quad \text{at home and} \quad r_x(z) = A \bar{\tau}^{1-\sigma} \left[\frac{z}{\phi(z)} \right]^{\sigma-1} \quad \text{abroad,}$$

and earns operating profits r/σ in each market.

Step 2: free entry. A firm serves a market when its operating profits cover the fixed cost of serving it, f_d or f_x (either may be zero). Free entry requires

$$F(A, \bar{\tau}) \equiv \int \max \left\{ \frac{r_d(z)}{\sigma} - f_d, 0 \right\} dG(z) + \int \max \left\{ \frac{r_x(z)}{\sigma} - f_x, 0 \right\} dG(z) = f_e.$$

The integrands are continuous and equal to zero at the boundaries of the sets of active firms, so changes in those sets do not affect F to a first order, and

$$\frac{\partial F}{\partial \ln A} = \frac{X_d + X_x}{\sigma}, \quad \frac{\partial F}{\partial \ln \bar{\tau}} = -\frac{(\sigma - 1) X_x}{\sigma},$$

with X_d and X_x the integrals of r_d and r_x over the active firms. Since f_e is fixed,

$$d \ln A = (\sigma - 1) \frac{X_x}{X_d + X_x} d \ln \bar{\tau} = (\sigma - 1)(1 - \lambda) d \ln \bar{\tau}. \quad (\text{A.3})$$

Step 3: welfare. With CES preferences and income per capita equal to the wage, welfare per capita is $W = 1/\mathcal{P}$, and $d \ln A = (\sigma - 1) d \ln \mathcal{P}$ because L is fixed. By (A.3), $d \ln W = -d \ln \mathcal{P} = -(1 - \lambda) d \ln \bar{\tau}$, the first equality in (19); (A.2) gives the second.

Step 4: Pareto productivity. Let $G(z) = 1 - (b/z)^\theta$, with $\theta > \sigma - 1$, and let $f_d > 0$ be large enough that the domestic cutoff z_d , defined by $Az_d^{\sigma-1} = \sigma f_d$, exceeds b . Integrating r_d above z_d and using (A.1),

$$X_d = \frac{\sigma f_d \theta}{\theta - \sigma + 1} \left(\frac{b}{z_d} \right)^\theta, \quad \lambda = \frac{N X_d}{L}.$$

Hence $d \ln \lambda = d \ln N - \theta d \ln z_d$. The cutoff condition gives $d \ln z_d = -d \ln A / (\sigma - 1) = -d \ln \mathcal{P} = d \ln W$, so

$$d \ln W = -\frac{1}{\theta} \left(1 - \frac{d \ln N}{d \ln \lambda} \right) d \ln \lambda,$$

and, comparing with (19), $\varepsilon_T = \theta / (1 - d \ln N / d \ln \lambda)$.

Step 5: power-law profile. Let $\phi(z) = (z/z_0)^{-\delta}$, with $\theta > (1 + \delta)(\sigma - 1)$, and let $f_x > 0$ make the export cutoff z_x exceed b . Then $r_x(z) = \sigma f_x (z/z_x)^{(1+\delta)(\sigma-1)}$ for $z \geq z_x$, and

$$X_x = \frac{\sigma f_x \theta}{\theta - (1 + \delta)(\sigma - 1)} \left(\frac{b}{z_x} \right)^\theta.$$

The fixed costs paid per entrant, $f_d(b/z_d)^\theta$ and $f_x(b/z_x)^\theta$, are therefore shares $(\theta - \sigma + 1)/(\sigma\theta)$ of X_d and $[\theta - (1 + \delta)(\sigma - 1)]/(\sigma\theta)$ of X_x , so profits net of fixed costs are shares

$$\zeta_d = \frac{\sigma - 1}{\sigma\theta} \quad \text{and} \quad \zeta_x = (1 + \delta) \zeta_d$$

of domestic sales and exports. Free entry requires $\zeta_d X_d + \zeta_x X_x = f_e$; with (A.1), $N = \bar{\zeta} L / f_e$, where $\bar{\zeta} = \lambda \zeta_d + (1 - \lambda) \zeta_x$. Since ζ_d and ζ_x are constants,

$$\frac{d \ln N}{d \ln \lambda} = \frac{\lambda (\zeta_d - \zeta_x)}{\bar{\zeta}} = -\frac{\lambda \delta}{1 + \delta(1 - \lambda)},$$

and Step 4 gives $\varepsilon_T = \theta [1 + \delta(1 - \lambda)] / (1 + \delta)$. □

A.2 Proof of Proposition 2

Step 1: demand. Each consumer has utility $U = \int_{\omega \in \mathcal{V}} u(q_\omega) d\omega$, with $u(0) = 0$, $u' > 0$, $u'' < 0$ and $u'(0) < \infty$; ACDR's generalized CES demand with $\beta = 0$ has this form, with $u(q) = (q + \alpha)^{1-1/\sigma} - \alpha^{1-1/\sigma}$ in the quantitative analysis. Let χ be the marginal utility of income and $P \equiv u'(0)/\chi$ the choke price. A variety with price p is demanded in quantity $q = D(p/P)$ per consumer, where $D(t)$ solves $u'(D) = t u'(0)$ for $t < 1$ and $D(t) = 0$ for $t \geq 1$. The set $\mathcal{V}$ consists of the domestic varieties of home entrants and the imported varieties of foreign entrants; by symmetry, both countries have N entrants and the same choke price.

Step 2: pricing. A firm with marginal cost c maximizes $(p - c) D(p/P)$. Its markup depends only on its relative efficiency $v \equiv P/c$, $p = \mu(v) c$, and it sells if and only if $v > 1$. With no fixed costs, the marginal firm in each market has $v = 1$ and sells nothing. Differentiating the pricing rule,

$$d \ln p = [1 - \rho(v)] d \ln c + \rho(v) d \ln P, \quad \rho(v) \equiv \frac{d \ln \mu}{d \ln v}. \quad (\text{A.4})$$

Domestic sales have $c = 1/z$, so $d \ln c = 0$. Exports have $c = \bar{\tau} \phi(z)/z$, so $d \ln c = d \ln \bar{\tau}$ for every exporter, because ϕ is fixed.

Step 3: welfare. Let e be the minimum expenditure that a consumer needs to reach utility U , given the prices and the set of varieties. Indexing domestic and imported varieties by the productivity of their producers,

$$e = \min_{q_d(\cdot), q_x(\cdot)} \sum_{m \in \{d, x\}} N \int_{Z_m} p_m(z) q_m(z) dG(z) \quad \text{s.t.} \quad \sum_{m \in \{d, x\}} N \int_{Z_m} u(q_m(z)) dG(z) \geq U,$$

where Z_d and Z_x are the sets of firms that sell at home and abroad. Let ν be the multiplier on the constraint; for varieties with positive demand, $\nu u'(q) = p$, so $\nu = 1/\chi$. By the envelope theorem, at fixed U : (i) $\partial e / \partial p_m(z) = N q_m(z) g(z)$ (Shephard's lemma); (ii) moving the boundary of Z_m has no effect, because the marginal varieties have $q = 0$ and therefore $p q - \nu u(q) = 0$; and (iii)

$$\frac{\partial e}{\partial N} = \sum_m \int_{Z_m} [p_m q_m - \nu u(q_m)] dG = -\frac{CS}{N}, \quad CS \equiv N \sum_m \int_{Z_m} \int_0^{q_m} [\nu u'(s) - p_m] ds dG,$$

where CS is the consumer surplus of a consumer, measured in units of income. Each consumer's income is the wage, which is fixed, so the equivalent variation satisfies, to a first

order, $d \ln W = -de/e$, where de is the change in e caused by the changes in prices and varieties at the initial utility.⁴ Hence

$$d \ln W = -\lambda \int s_d(z) d \ln p_d(z) dz - (1 - \lambda) \int s_x(z) d \ln p_x(z) dz + \bar{S} d \ln N, \quad (\text{A.5})$$

where $s_m(z) = p_m(z)q_m(z)g(z)/\int_{Z_m} p_m q_m dG$ is the density of sales across firms in market m and $\bar{S} = CS/e = \lambda S_d + (1 - \lambda)S_x$.

Step 4: aggregation. By Step 2, $\int s_d d \ln p_d dz = \rho_d d \ln P$ and $\int s_x d \ln p_x dz = (1 - \rho_x) d \ln \bar{\tau} + \rho_x d \ln P$, with $\rho_m = \int s_m(z) \rho(v_m(z)) dz$. Substituting in (A.5),

$$d \ln W = -(1 - \lambda)(1 - \rho_x) d \ln \bar{\tau} - \bar{\rho} d \ln P + \bar{S} d \ln N. \quad (\text{A.6})$$

Step 5: the welfare coefficient. By (A.2), $d \ln \bar{\tau} = d \ln \lambda / [(1 - \lambda) \varepsilon_T]$. Substituting this, $d \ln P = (d \ln P / d \ln \bar{\tau}) d \ln \bar{\tau}$ and $d \ln N = (d \ln N / d \ln \lambda) d \ln \lambda$ in (A.6),

$$d \ln W = - \left[\frac{1 - \rho_x}{\varepsilon_T} + \frac{\bar{\rho}}{(1 - \lambda) \varepsilon_T} \frac{d \ln P}{d \ln \bar{\tau}} - \bar{S} \frac{d \ln N}{d \ln \lambda} \right] d \ln \lambda = -\Omega d \ln \lambda,$$

with Ω as in (21).

Step 6: entry and the choke price. A firm with relative efficiency v earns profits $LP\pi(v)$, with $\pi(v) \equiv q(v)[\tilde{p}(v) - 1/v]$, $\tilde{p}(v) \equiv \mu(v)/v$ and $q(v) = D(\tilde{p}(v))$. By the envelope theorem, profits fall with the marginal cost at the rate Lq ; since $v = P/c$, this gives $\pi'(v) = q(v)/v^2$, and $\pi(1) = 0$. Let Π_d and Π_x be the expected domestic and export profits of an entrant, so that $\Pi_m = \zeta_m X_m$. Free entry requires $\Pi_d + \Pi_x = f_e$; with (A.1), $N = \bar{\zeta} L / f_e$. Since $v_d = Pz$ and $v_x = Pz / [\bar{\tau}\phi(z)]$, and the boundaries of Z_d and Z_x do not contribute because $\pi(1) = 0$,

$$\begin{aligned} \frac{\partial \Pi_m}{\partial \ln P} &= L \int_{Z_m} P [\pi(v_m) + v_m \pi'(v_m)] dG = X_m, \\ \frac{\partial \Pi_x}{\partial \ln \bar{\tau}} &= -L \int_{Z_x} P v_x \pi'(v_x) dG = -(1 - \zeta_x) X_x. \end{aligned}$$

because $P[\pi(v) + v\pi'(v)] = P\tilde{p}(v)q(v)$ is sales per consumer and $Pv\pi'(v) = Pq(v)/v$ is variable cost per consumer. Differentiating free entry, $(X_d + X_x) d \ln P = (1 - \zeta_x) X_x d \ln \bar{\tau}$, so $d \ln P / d \ln \bar{\tau} = (1 - \lambda)(1 - \zeta_x)$, which completes (22). $\square$

⁴Let V be indirect utility and $W = e(\mathbf{p}^0, \mathcal{V}^0, V)$ money-metric welfare at the initial prices and varieties. Differentiating $e(\mathbf{p}, \mathcal{V}, V) = 1$ gives $e_U dV = -de$, and $dW = e_U dV$ at the initial point, where $W = 1$.

A.3 Proof of Corollary 1

Step 1: relative efficiency. With $\phi(z) = (z/z_0)^{-\delta}$, a firm's relative efficiency is $v_d = Pz$ at home and $v_x = Pz^{1+\delta}/(\bar{\tau}z_0^\delta)$ abroad, and it sells in a market if and only if $v > 1$. Under $G(z) = 1 - (b/z)^\theta$, with b below both cutoffs, the measures of sellers per entrant are $n_d = (bP)^\theta$ and $n_x = (b/z_x)^\theta$, with $z_x = (\bar{\tau}z_0^\delta/P)^{1/(1+\delta)}$, and, among exporters,

$$\begin{aligned}\Pr(v_x \geq t) &= \frac{\Pr(z \geq z_x t^{1/(1+\delta)})}{\Pr(z \geq z_x)} \\ &= t^{-\theta/(1+\delta)}, \quad t \geq 1.\end{aligned}$$

Relative efficiency is thus Pareto on $[1, \infty)$, with shape θ among domestic sellers and $\theta_x \equiv \theta/(1+\delta)$ among exporters, whatever P and $\bar{\tau}$. Let E_k denote the expectation under a Pareto distribution with shape k on $[1, \infty)$.

Step 2: profit shares. For $k \in \{\theta, \theta_x\}$, with $k > \sigma - 1$ so that aggregate sales are finite, integration by parts and $\pi'(v) = q(v)/v^2$ (Step 6 of the previous proof) give

$$E_k[\pi] = \int_1^\infty \pi(v) k v^{-k-1} dv = \int_1^\infty \pi'(v) v^{-k} dv = \frac{1}{k} E_k\left[\frac{q}{v}\right].$$

The boundary terms vanish: $\pi(1) = 0$, and since π is increasing and $E_k[\pi]$ is finite, $\pi(v) v^{-k} \leq \int_v^\infty \pi(s) k s^{-k-1} ds \rightarrow 0$. Aggregate profits are thus a share $1/k$ of variable costs and a share $1/(1+k)$ of sales, so $\zeta_d = 1/(1+\theta)$ and $\zeta_x = 1/(1+\theta_x) = (1+\delta)/(1+\theta+\delta)$.

Step 3: entry and the choke price. By (22), $N = \bar{\zeta}L/f_e$. Since ζ_d and ζ_x are constants, $d \ln N = (\zeta_d - \zeta_x) d\lambda/\bar{\zeta}$, which gives (23) after substituting ζ_d and ζ_x . Also by (22),

$$\frac{d \ln P}{d \ln \bar{\tau}} = (1-\lambda)(1-\zeta_x) = (1-\lambda) \frac{\theta}{1+\theta+\delta},$$

which is (24).

Step 4: trade elasticity. By Step 1,

$$X_d = K_d P^{1+\theta}, \quad X_x = K_x \bar{\tau}^{-\theta_x} P^{1+\theta_x},$$

where K_d and K_x are constants. Hence

$$\frac{1-\lambda}{\lambda} = \frac{X_x}{X_d} = \frac{K_x}{K_d} \bar{\tau}^{-\theta_x} P^{\theta_x-\theta},$$

and therefore

$$\varepsilon_T = \theta_x + (\theta - \theta_x) \frac{d \ln P}{d \ln \bar{\tau}},$$

which is (25) because $\theta - \theta_x = \theta \delta / (1 + \delta)$.

Step 5: markups and consumer surplus. Let $Pcs(v)$ be the consumer surplus that a consumer obtains from a variety with relative efficiency v . Then $\rho_m = E_k[\rho \tilde{p}q] / E_k[\tilde{p}q]$ and $S_m = E_k[cs] / E_k[\tilde{p}q]$, with $k = \theta$ for domestic sales and $k = \theta_x$ for exports, so both depend only on k . Consumer surplus falls with the price at the rate q , and $d\tilde{p}/dv = -(1 - \rho) \tilde{p}/v$, so $cs'(v) = (1 - \rho) \tilde{p}q/v$, with $cs(1) = 0$. Integrating by parts as in Step 2, $E_k[cs] = E_k[(1 - \rho) \tilde{p}q]/k$, so $S_m = (1 - \rho_m)/k$: $S_d = (1 - \rho_d)/\theta$ and $S_x = (1 + \delta)(1 - \rho_x)/\theta$.

Step 6: the ACDR benchmark. With $\delta = 0$, $\theta_x = \theta$, so $\zeta_d = \zeta_x$, $d \ln N / d \ln \lambda = 0$, $\varepsilon_T = \theta$, and $\rho_d = \rho_x \equiv \rho$. Moreover,

$$\frac{d \ln P}{d \ln \bar{\tau}} = \frac{(1 - \lambda)\theta}{1 + \theta}.$$

Substituting in (21) gives

$$\Omega = \frac{1 - \rho}{\theta} + \frac{\rho}{1 + \theta} = \frac{1 - \rho/(1 + \theta)}{\theta}.$$

□

A.4 Proof of Proposition 3

Part (i). In Step 2 of the proof of Proposition 1, replace the common change $d \ln \bar{\tau}$ by firm specific changes $d \ln \tau_x(z)$. The boundaries of the sets of active firms still do not contribute, and $\partial F / \partial \ln \tau_x(z) = -(\sigma - 1) r_x(z) g(z) / \sigma$ for each exporter, so

$$d \ln A = (\sigma - 1) \frac{\int r_x(z) d \ln \tau_x(z) dG(z)}{X_d + X_x} = (\sigma - 1)(1 - \lambda) E_s[d \ln \tau_x],$$

and Step 3 gives (27).

Part (ii). Steps 1 to 4 of the proof of Proposition 2 hold with $d \ln c = d \ln \tau_x(z)$ for exporter z , so $\int s_x d \ln p_x dz = E_s[(1 - \rho) d \ln \tau_x] + \rho_x d \ln P$, and (A.5) gives (28). In Step 6, $\partial \Pi_x / \partial \ln \tau_x(z) = -L P v \pi'(v) g(z) = -L P [q(v)/v] g(z)$, the variable cost of the exports of

firm z . Differentiating free entry,

$$(X_d + X_x) d \ln P = L \int_{Z_x} P \frac{q(v_x)}{v_x} d \ln \tau_x(z) dG(z) = (1 - \zeta_x) X_x E_c[d \ln \tau_x],$$

which gives (29); $N = \bar{\zeta}L/f_e$ follows from free entry and (A.1) as before. $\square$

Numerical check. Table A.1 scales the heterogeneous shock of Table 6 by factors from 1 to 0.001 and compares it with the uniform shock with the same trade weighted mean. In the generalized CES economy the welfare gap, relative to the welfare change of the uniform shock, converges to the first order value implied by (28); in a CES economy with the same profile it vanishes, as part (i) implies. The table reports the gap relative to the welfare change of U because both losses vanish as the shock is scaled down; at the observed scale, the absolute difference is 0.027 percentage points.

Table A.1: Numerical check of Proposition 3

Scale of the shock	Mean shock (log pts.)	Welfare of U (%)	Welfare gap, H minus U (% of $ U $ loss)	
			Generalized CES	CES
1	11.4	−0.6019	4.42	1.2373
0.1	1.14	−0.0795	3.12	0.0989
0.01	0.114	−0.0082	2.89	0.0105
0.001	0.0114	−0.0008	2.86	0.0019
First order, eq. (28)			2.86	0

Notes: Heterogeneous shock of Table 6 (2019 to 2021–22), multiplied by the factor in the first column, against a uniform shock with the same trade weighted mean (U). Generalized CES: calibrated economy with the saturating profile. CES: economy with $\theta = 5$, the same profile and fixed costs of serving each market, calibrated to the import share (its trade elasticity is 4.1; with these fixed costs a trade elasticity of 5 is not reached while the domestic cutoff stays above the Pareto lower bound). The welfare of U is for the generalized CES economy. First order prediction: linearized welfare change from equation (28) for each shock; zero under CES by equation (27).

B Empirical Robustness

The tables in this appendix report the fuller set of empirical specifications summarized in Sections 3–4.

Shipment value on both sides of the regression. The FOB value of a shipment is the denominator of its freight rate and, in Table 2, also a regressor. Table B.5 examines whether this overlap drives the conditional size gradient, using previous year size throughout. The overlap is not a problem in itself: once log value is a regressor, replacing $\ln(\text{Freight}/\text{FOB})$ with $\ln \text{Freight}$ as the dependent variable changes only the value coefficient, by exactly 100, so columns (3) and (4) report the same size coefficient. What could bias the gradient is error in declared values, which enters the dependent variable and the value regressor with opposite signs (the division bias of Borjas, 1980). Columns (6)–(8) address errors specific to a shipment. Column (6) instruments the shipment’s value with the firm’s mean log value on its other shipments in the year, column (7) replaces the shipment’s own weight and value with those means, and column (8) drops extreme unit values. The size coefficient lies between -0.16 and -0.20 . Classical errors in declared values would make the OLS value coefficient more negative than the IV one. The opposite holds, which suggests that the instrument also picks up the firm’s persistent shipment scale, so column (6) also nets that component out of the size coefficient. Because the instrument may also proxy for other persistent features of the firm’s shipping, such as its contracts or relationships with carriers, column (6) should be read as a sensitivity check rather than as a causal estimate.

Systematic under declaration by small importers would survive these checks, but freight per kilogram in column (5) does not use declared values at all. Since column (5) is equivalent to dropping value from column (4), the gap between their size coefficients is the value coefficient in column (4) times the elasticity of declared value with respect to firm size, conditional on the other regressors (0.029). If the higher declared unit values of larger importers reflected under declaration by smaller ones rather than differences in the goods shipped, column (5) would be the relevant estimate. It is 58 percent of column (4) and precisely estimated. Finally, monthly size includes the shipment itself. Among firm-months with other imports, removing the shipment from monthly size lowers the size coefficient from -0.45 to -0.33 without shipment controls and from -0.26 to -0.21 with them, one reason to emphasize predetermined size.

Table B.1: Firm size and transport costs: all maritime routes

	(1)	(2)	(3)
Total imports (log)	−0.660*** (0.069)	−0.693*** (0.068)	−1.164*** (0.031)
Experience (years importing)	−0.292*** (0.012)	−0.189*** (0.024)	0.480*** (0.067)
Weight (log)	0.436*** (0.032)	0.435*** (0.032)	0.142*** (0.043)
Route FE	Yes	Yes	Yes
Month-year FE	No	Yes	Yes
Firm FE	No	No	Yes
Observations	4,466,633	4,466,633	4,438,696
Firms	71,738	71,738	43,946
R^2	0.217	0.223	0.459

Notes: Dependent variable: $100 \times \ln(1 + \text{freight}/\text{FOB})$, approximately the ad valorem freight rate in percentage points. Maritime imports, 2007–2017, excluding transactions with imputed freight (5% of FOB). Total imports are the firm’s FOB imports in the month, all origins. All transactions. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.2: Carriers and the size gradient

	(1)	(2)	(3)	(4)	(5)
Total imports (log)	-0.551*** (0.028)	-0.554*** (0.027)	-0.741*** (0.035)	-0.765*** (0.034)	-1.159*** (0.048)
Transactions with the carrier (log)			0.292*** (0.032)	0.305*** (0.030)	-0.169*** (0.023)
Total transactions of the firm (log)					1.072*** (0.058)
Number of carriers used (log)					-0.553*** (0.089)
Experience (years importing)	-0.097*** (0.025)	-0.097*** (0.024)	-0.107*** (0.025)	-0.133*** (0.022)	-0.164*** (0.024)
Weight (log)	-0.109*** (0.027)	-0.117*** (0.027)	-0.088*** (0.027)	-0.135*** (0.026)	-0.082*** (0.027)
Product FE	Yes	Yes	Yes	Yes	Yes
Route and month-year FE	Yes	Yes	Yes	—	—
Carrier FE	No	Yes	Yes	—	—
Carrier $\times$ route $\times$ month-year FE	No	No	No	Yes	Yes
Observations	3,373,208	3,373,208	3,373,208	3,005,791	3,005,791
Firms	60,127	60,127	60,127	55,584	55,584
R^2	0.423	0.425	0.426	0.547	0.549

Notes: Dependent variable: $100 \times \ln(1 + \text{freight}/\text{FOB})$, approximately the ad valorem freight rate in percentage points. Maritime imports, 2007–2017, excluding transactions with imputed freight (5% of FOB). Total imports are the firm’s FOB imports in the month, all origins. Single product declarations with an identified carrier (1,098 carriers). Transactions, carriers and total transactions are counted over the firm’s maritime imports in the year. Columns (1)–(3) use the same sample; columns (4)–(5) drop singletons of the carrier $\times$ route $\times$ month-year fixed effects. In columns (4)–(5), carrier $\times$ route $\times$ month-year fixed effects absorb the carrier, route and month-year fixed effects. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.3: Carriers and the size gradient: Shanghai–San Antonio

	(1)	(2)	(3)	(4)	(5)
Total imports (log)	−0.486*** (0.075)	−0.468*** (0.072)	−0.508*** (0.096)	−0.546*** (0.089)	−0.826*** (0.150)
Transactions with the carrier (log)			0.054 (0.069)	0.098 (0.062)	−0.097 (0.071)
Total transactions of the firm (log)					0.541*** (0.177)
Number of carriers used (log)					−0.132 (0.206)
Experience (years importing)	−0.113* (0.066)	−0.093 (0.062)	−0.094 (0.062)	−0.057 (0.053)	−0.092 (0.058)
Weight (log)	0.039 (0.071)	0.017 (0.064)	0.024 (0.067)	0.013 (0.064)	0.054 (0.069)
Product FE	Yes	Yes	Yes	Yes	Yes
Month-year FE	Yes	Yes	Yes	–	–
Carrier FE	No	Yes	Yes	–	–
Carrier × month-year FE	No	No	No	Yes	Yes
Observations	66,563	66,563	66,563	64,963	64,963
Firms	5,431	5,431	5,431	5,257	5,257
R^2	0.409	0.417	0.417	0.473	0.475

Notes: Dependent variable: $100 \times \ln(1 + \text{freight}/\text{FOB})$, approximately the ad valorem freight rate in percentage points. Maritime imports from Shanghai to San Antonio, 2007–2017, excluding transactions with imputed freight. Single-product declarations with an identified carrier. In columns (4)–(5), carrier × month-year fixed effects absorb the carrier and month-year fixed effects. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.4: Shipment size and the weighting of importers: Shanghai–San Antonio

	(1) Each shipment weighs the same	(2) Each shipment weighs the same	(3) Each firm-year weighs the same	(4) Each firm-year weighs the same
Firm size, previous year (log)	−0.333*** (0.050)	−0.202*** (0.043)	−0.328*** (0.042)	−0.058 (0.039)
Shipment weight and value (log)	No	Yes	No	Yes
Product and month-year FE	Yes	Yes	Yes	Yes
Observations	60,578	60,578	60,578	60,578
Firms	4,377	4,377	4,377	4,377

Notes: Dependent variable: $100 \times \ln(1 + \text{freight}/\text{FOB})$, approximately the ad valorem freight rate in percentage points. Maritime imports from Shanghai to San Antonio, 2007–2017, single product declarations, excluding transactions with imputed freight, importers that also imported in the previous year. Firm size is the importer’s FOB imports in the previous year, all origins. In columns (3)–(4) each transaction is weighted by the inverse of the importer’s number of transactions in the year. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.5: Shipment value on both sides of the regression

	(1) Freight rate	(2) Freight rate	(3) $\ln(F/\text{FOB})$	(4) $\ln F$	(5) $\ln(F/\text{kg})$	(6) IV	(7) Typical shipment Freight rate	(8) Trimmed unit values
Firm size, previous year (log)	−0.347*** (0.020)	−0.220*** (0.020)	−2.582*** (0.275)	−2.582*** (0.275)	−1.506*** (0.275)	−0.166*** (0.026)	−0.204*** (0.027)	−0.162*** (0.019)
Weight (log)	−0.165*** (0.027)	3.251*** (0.070)	51.040*** (0.687)	51.040*** (0.687)	−20.004*** (0.512)	4.310*** (0.336)		3.297*** (0.073)
Shipment FOB value (log)		−4.354*** (0.094)	−63.089*** (0.846)	36.911*** (0.846)		−5.693*** (0.433)		−4.363*** (0.096)
Mean log weight, other shipments							1.578*** (0.084)	
Mean log value, other shipments							−2.545*** (0.132)	
Product, route and month-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F (Kleibergen–Paap)						333.3		
Observations	3,046,124	3,046,124	3,046,124	3,046,124	3,046,124	3,015,956	3,015,956	2,758,067
Firms	32,378	32,378	32,378	32,378	32,378	23,014	23,014	30,289
R^2	0.424	0.523	0.637	0.876	0.658	–	0.445	0.558

Notes: Dependent variables are multiplied by 100. The freight rate is $100\ln(1 + F/\text{FOB})$, where F denotes freight charges; columns (3)–(5) use the logs of F/FOB , F and F/kg instead, so their coefficients are on a different scale from those in the other columns. Single product maritime imports, 2007–2017, excluding transactions with imputed freight and the 35 with zero freight, for which the logs in columns (3)–(5) are undefined. One of these 35 is a singleton that Table 2 already drops, so column (2) has 34 fewer observations than Table 2, column (3), and otherwise reproduces it. Firm size is the importer’s FOB imports in the previous year, all origins. All columns control for importing experience. Column (6) instruments log shipment value with the mean log value of the firm’s other single product maritime shipments in the same year; OLS on that sample gives a size coefficient of -0.206 (0.021). Column (7) uses the same sample and replaces the shipment’s own weight and value with the mean logs over those other shipments. Column (8) drops shipments whose value per kilogram lies outside the 5th–95th percentiles of their product-year. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.6: Freight costs and imputation by importer size

Size quintile	Imputed insurance (% of transactions)	Imputed freight (% of transactions)	Insurance/FOB (mean, %)	Freight/FOB (mean, %)	Transactions
1 (smallest)	56.6	1.3	1.90	22.0	53,333
2	42.6	1.0	1.34	13.9	85,752
3	37.7	1.0	1.18	11.5	172,861
4	30.9	1.4	1.00	9.8	420,757
5 (largest)	8.8	0.9	0.41	7.0	3,797,954

Notes: Maritime imports, 2007–2017, all transactions. Importers are assigned to quintiles of their total imports (FOB, all origins) within each year. Imputed insurance: insurance equal to 2% of FOB; imputed freight: freight equal to 5% of FOB, the values Chilean Customs imputes when insurance or freight are not documented.

Table B.7: Alternative measures of firm size

	(1)	(2)	(3)	(4)	(5)
	FOB, month	CIF, month	FOB, year, excl. shipment	FOB, previous year	FOB, month, no weight
Firm size (log)	-0.551*** (0.028)	-0.493*** (0.025)	-0.398*** (0.021)	-0.347*** (0.020)	-0.566*** (0.027)
Experience (years importing)	-0.097*** (0.025)	-0.115*** (0.025)	-0.073*** (0.027)	-0.046 (0.029)	-0.096*** (0.024)
Weight (log)	-0.109*** (0.027)	-0.121*** (0.027)	-0.142*** (0.027)	-0.165*** (0.027)	
Product FE	Yes	Yes	Yes	Yes	Yes
Month-year FE	Yes	Yes	Yes	Yes	Yes
Route FE	Yes	Yes	Yes	Yes	Yes
Observations	3,373,606	3,373,606	3,338,064	3,046,158	3,373,608
Firms	60,158	60,158	39,634	32,378	60,158
R^2	0.423	0.421	0.425	0.424	0.423

Notes: Dependent variable: $100 \times \ln(1 + \text{freight}/\text{FOB})$, approximately the ad valorem freight rate in percentage points. Maritime imports, 2007–2017, excluding transactions with imputed freight (5% of FOB). Single-product declarations. The size row reports the coefficient on each measure; column (1) is the baseline. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.8: Alternative measures of transport costs

	(1)	(2)	(3)	(4)
	$\ln(1 + F/\text{FOB})$	Incl. imputed	$\ln(\text{CIF}/\text{FOB})$	Freight/CIF
Total imports (log)	-0.551*** (0.028)	-0.547*** (0.028)	-0.648*** (0.029)	-0.437*** (0.022)
Experience (years importing)	-0.097*** (0.025)	-0.096*** (0.024)	-0.118*** (0.025)	-0.073*** (0.019)
Weight (log)	-0.109*** (0.027)	-0.098*** (0.026)	-0.188*** (0.028)	-0.052** (0.021)
Product FE	Yes	Yes	Yes	Yes
Month-year FE	Yes	Yes	Yes	Yes
Route FE	Yes	Yes	Yes	Yes
Observations	3,373,606	3,414,780	3,373,606	3,373,606
Firms	60,158	60,556	60,158	60,158
R^2	0.423	0.421	0.426	0.453

Notes: Dependent variables: (1) $100 \times \ln(1 + \text{freight}/\text{FOB})$; (2) the same, including transactions with imputed freight; (3) $100 \times \ln(\text{CIF}/\text{FOB})$, which also includes insurance; (4) freight/CIF in percentage points. Maritime imports, 2007–2017, single product declarations. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.9: Firm size and transport costs by period

	(1) 2007–2017	(2) 2019–2024	(3) 2020–2022
Total imports (log)	−0.551*** (0.028)	−0.668*** (0.014)	−0.895*** (0.022)
Experience (years importing)	−0.097*** (0.025)	−0.001 (0.005)	−0.000 (0.008)
Weight (log)	−0.109*** (0.027)	0.270*** (0.012)	0.524*** (0.017)
Product FE	Yes	Yes	Yes
Month-year FE	Yes	Yes	Yes
Route FE	Yes	Yes	Yes
Observations	3,373,606	2,311,214	1,188,614
Firms	60,158	23,740	17,303
R^2	0.423	0.418	0.445

Notes: Dependent variable: $100 \times \ln(1 + \text{freight}/\text{FOB})$, approximately the ad valorem freight rate in percentage points. Maritime imports, single product declarations, excluding transactions with imputed freight. Importer identifiers change in 2018, which is excluded. Standard errors clustered by firm in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C Additional Quantitative Results

C.1 Exporter sales distribution

Table C.1 reports the exporter sales moments discussed in Section 7. The power law profile is almost identical to ACDR once the aggregate trade elasticity is held fixed; the saturating profile compresses the lower tail.

Table C.1: Export sales relative to the mean, by percentile of exporters

	Percentile of exporters				
	5	25	50	75	95
French data	0.006	0.025	0.094	0.401	3.363
Constant costs (ACDR)	0.027	0.160	0.417	0.969	3.243
Power law profile	0.027	0.160	0.416	0.967	3.241
Saturating profile	0.064	0.262	0.524	1.050	3.090

Notes: Export sales relative to mean export sales. French moments are those reported in the ACDR working paper calibration used for the demand parameters in the quantitative exercise.

C.2 Welfare robustness

Table C.2 recalibrates the model using alternative moments for the freight cost profile. The results bound the additional local welfare effect and show how much of it survives after conditioning on shipment size.

Table C.2: Welfare robustness to alternative profile moments

Profile / moment	Micro estimate	θ	Exporters (%)	Ω	Ω vs. ACDR (%)	$\bar{\tau} \times 0.8$
Power law, previous-year rank	$r = 0.0048$	5.116	7.08	0.1960	2.1	2.64
Power law, IV previous-year rank	$r = 0.0050$	5.120	7.07	0.1961	2.2	2.65
Power law, CIF/FOB	$r = 0.0086$	5.210	6.80	0.1991	3.8	2.68
Power law, all maritime routes	$r = 0.0090$	5.219	6.77	0.1994	3.9	2.69
Power law, weighted by shipments	$r = 0.0042$	5.102	7.13	0.1955	1.9	2.64
Power law, net shipment size (shipment-weighted)	$r = 0.0019$	5.045	7.32	0.1935	0.8	2.61
Power law, net shipment size (equal firm-year weights)	$r = -0.0026$	4.939	7.69	0.1896	-1.2	2.56
Saturating, previous-year rank	$\kappa = .027, \psi = 5.26$	5.326	6.59	0.2008	4.7	2.82
Saturating, 10 bins	$\kappa = .041, \psi = 6.68$	5.528	6.15	0.2056	7.2	2.88
Saturating, 50 bins	$\kappa = .047, \psi = 9.42$	5.630	5.95	0.2079	8.3	2.91
Saturating, exclude smallest 2%	$\kappa = .031, \psi = 5.39$	5.381	6.47	0.2022	5.4	2.84

Notes: Every specification is recalibrated to $\varepsilon_T = 5$ and imports/expenditure of 6.95 percent. Ω is the local welfare sensitivity. The two “net shipment size” rows control for the firm’s average shipment weight and value and differ only in whether observations are weighted by shipments or equally by firm-year. The row with equal firm-year weights does not impose the baseline restriction $\phi'(z) \leq 0$: its slightly negative estimate implies $\delta < 0$, so larger firms face marginally higher export costs. It is reported as a lower bound. The last column is the exact welfare gain, in percent, from $\bar{\tau} \times 0.8$.